# Decent Coaching Classes

## Sarvadnya

Class VIII | Maths Worksheet | Algebraic Expressions, Factorisation & Linear Equations in One Variable

### Section A: Algebraic Expressions and Identities

**1.** Classify each expression as a monomial, binomial or trinomial:

(i) $7xy$

(ii) $4x^2 - 9$

(iii) $3a + 2b - 5$

(iv) $6p^2q$

**2.** Add: $5x^2 - 3xy + 2y^2$ and $-2x^2 + 7xy - 4y^2$.

**3.** Subtract $3a^2 - 4ab + 5b^2$ from $7a^2 + 2ab - 3b^2$.

**4.** Multiply and write the product in simplest form: $(2x - 3)(x^2 + 4x - 5)$.

**5.** Using a suitable identity:

(i) Expand $(3x + 2y)^2$.

(ii) Find the value of $103 \times 97$.

### Section B: Factorisation

**6.** Factorise: $12x^2y - 18xy^2$.

**7.** Factorise by regrouping the terms: $6ab - b^2 + 12ac - 2bc$.

**8.** Factorise using identities:

(i) $9x^2 - 24x + 16$

(ii) $49a^2 - 36b^2$

**9.** Factorise:

(i) $x^2 + 9x + 20$

(ii) $y^2 - 7y - 18$

**10.** Divide:

(i) $28x^3 - 14x^2 + 7x$ by $7x$

(ii) $(x^2 + 7x + 10)$ by $(x + 2)$

### Section C: Linear Equations in One Variable

**11.** Solve and check your answer: $5x - 7 = 3x + 9$.

**12.** Solve: $3(x - 2) = 2(x + 1) + 5$.

**13.** Solve: $(2x + 1)/3 = (x + 4)/2$.

**14.** Solve the equation reducible to linear form: $(x + 1)/(2x + 3) = 3/8$.

**15.** The length of a rectangle is 5 cm more than twice its breadth. If its perimeter is 52 cm, find the length and breadth.

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Answer Key and Solutions

### Section A

**1.** (i) Monomial (ii) Binomial (iii) Trinomial (iv) Monomial

**2.** $(5x^2 - 2x^2) + (-3xy + 7xy) + (2y^2 - 4y^2)$
$= 3x^2 + 4xy - 2y^2$

**3.** $(7a^2 + 2ab - 3b^2) - (3a^2 - 4ab + 5b^2)$
$= 7a^2 - 3a^2 + 2ab + 4ab - 3b^2 - 5b^2$
$= 4a^2 + 6ab - 8b^2$

**4.** $2x(x^2 + 4x - 5) - 3(x^2 + 4x - 5)$
$= 2x^3 + 8x^2 - 10x - 3x^2 - 12x + 15$
$= 2x^3 + 5x^2 - 22x + 15$

**5.** (i) $(a + b)^2 = a^2 + 2ab + b^2 \rightarrow (3x)^2 + 2(3x)(2y) + (2y)^2 = 9x^2 + 12xy + 4y^2$
(ii) $103 \times 97 = (100 + 3)(100 - 3) = 100^2 - 3^2 = 10000 - 9 = 9991$

### Section B

**6.** HCF of the terms = $6xy$
$12x^2y - 18xy^2 = 6xy(2x - 3y)$

**7.** $6ab - b^2 + 12ac - 2bc = b(6a - b) + 2c(6a - b)$
$= (6a - b)(b + 2c)$

**8.** (i) $9x^2 - 24x + 16 = (3x)^2 - 2(3x)(4) + 4^2 = (3x - 4)^2$
(ii) $49a^2 - 36b^2 = (7a)^2 - (6b)^2 = (7a - 6b)(7a + 6b)$

**9.** (i) Two numbers with sum 9 and product 20 are 4 and 5: $x^2 + 9x + 20 = (x + 4)(x + 5)$
(ii) Two numbers with sum −7 and product −18 are −9 and 2: $y^2 - 7y - 18 = (y - 9)(y + 2)$

**10.** (i) $7x(4x^2 - 2x + 1) \div 7x = 4x^2 - 2x + 1$
(ii) $x^2 + 7x + 10 = (x + 2)(x + 5)$, so the quotient = $x + 5$

### Section C

**11.** $5x - 3x = 9 + 7 \rightarrow 2x = 16 \rightarrow x = 8$
Check: LHS = $5(8) - 7 = 33$; RHS = $3(8) + 9 = 33$ ✓

**12.** $3x - 6 = 2x + 2 + 5 \rightarrow 3x - 6 = 2x + 7 \rightarrow x = 13$

**13.** Cross-multiplying: $2(2x + 1) = 3(x + 4)$
$4x + 2 = 3x + 12 \rightarrow x = 10$

**14.** Cross-multiplying: $8(x + 1) = 3(2x + 3)$
$8x + 8 = 6x + 9 \rightarrow 2x = 1 \rightarrow x = 1/2$

**15.** Let the breadth = b cm, so the length = $(2b + 5)$ cm.
Perimeter = $2(\text{length} + \text{breadth}) = 2(2b + 5 + b) = 52$
$3b + 5 = 26 \rightarrow 3b = 21 \rightarrow b = 7$
Breadth = 7 cm, Length = $2(7) + 5 = 19$ cm